Besides the symplectic structure there are other possible choices which could underpin the design of numerical methods. For example, symplectic methods are volume preserving, but it is possible to construct volume preserving methods that are not symplectic. Similarly time-reversibility is a fundamental property of molecular dynamics models (due to the quadratic kinetic energy function) and could be used as a starting point for developing numerical schemes. We discuss these alternatives in this chapter. We also look at the performance of methods which are specifically designed to preserve the energy using a projection technique. The conclusion of this chapter is that symplectic methods are to be preferred in most cases; this observation is in agreement with current practice in molecular dynamics simulation. In molecular dynamics we typically have a restricted class of systems which are purely Hamiltonian, so geometric integration seems the natural starting point. A related concept to geometric integration is that of *variational integration* [395] which is in some ways more general (including some mechanical systems that are not in Hamiltonian form) but does not provide a useful backward error analysis (except in the same cases as geometric integration).

Geometric integration has been developed as a subject from around 1985 to the present. There remain many interesting open questions in this area. The discussion presented here is necessarily highly compressed and focussed to the issues directly relevant for molecular dynamics. For starting points to more comprehensive understanding of the topic, one may study the books [164, 227] or articles [121, 369].

3.1 The Modified Hamiltonian

We begin by asking the following question which takes us right to the heart of what is needed in molecular dynamics: how is the energy affected by numerical discretization?

Let us begin with an illustrative example. Consider the harmonic oscillator with frequency Ω which has Hamiltonian $H(q,p) = p^2/2 + \Omega^2 q^2/2$, and consider the adjoint symplectic Euler method

$$Q = q + hp, \qquad P = p - h\Omega^2 Q,$$

If $H(q,p) = \mathrm{E}$ then, for typical steps, we cannot expect $H(Q,P) = \mathrm{E}$. (Just insert the formulas for Q and P into the Hamiltonian and check that the value is not the same as $H(q,p)$.)

However, if we modify the Hamiltonian from $H(q,p) = p^2/2 + \Omega^2 q^2/2$ to

$$\tilde{H}(q,p) = \frac{p^2 + h\Omega^2 pq + \Omega^2 q^2}{2}, \tag{3.1}$$